

METIMMOX-1 Clinical Effectiveness Analysis Report

Biomarker-Guided Strategies for Selecting Immunotherapy Candidates in Metastatic MSS/pMMR Colorectal Cancer

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1 Overview

This report presents clinical effectiveness results for the METIMMOX-1 trial, comparing biomarker-guided treatment strategies for selecting immunotherapy candidates in metastatic MSS/pMMR colorectal cancer. The analysis compares four strategies: (1) Control (FLOX alone), (2) CRP-guided, (3) TLR-guided, and (4) TMB/BRAF-guided. Median survival times are derived from age- and sex-adjusted parametric survival models weighted by biomarker prevalence. Hazard ratios are estimated using Cox proportional hazards models with biomarker×treatment interactions, adjusted for age and sex. Life years and QALYs are calculated from the decision-analytic model. All outcomes are **undiscounted** (discount rate = 0%).

2 Results

2.1 Cox Proportional Hazards Models

Cox proportional hazards regression models were fitted with age, sex, treatment, and all three biomarker×treatment interaction terms.

2.2 Median Survival Times

2.3 Clinical Effectiveness Summary

Note: All outcomes are undiscounted (discount rate = 0%). Median survival times derived from parametric survival models weighted by biomarker prevalence. Hazard ratios (HR) from Cox proportional hazards models adjusted for age, sex, and all biomarker×treatment interactions. Life years and QALYs calculated from decision-analytic model. CI = Confidence Interval; LY = Life Years; QALY = Quality-Adjusted Life Years; PFS = Progression-Free Survival; OS = Overall Survival.

Table 1: Clinical Effectiveness Results (Undiscounted)

Strategy	Median PFS (months)	Median OS (months)	HR for PFS (95% CI)	HR for OS (95% CI)	Mean Life Years	Mean QALYs	Incremental LY vs. Control	Incremental QALYs vs. Control
Control (FLOX alone)	11.74	22.78	Reference	Reference	2.10	1.80	0.00	0.00
CRP-guided	13.12	24.17	1.52 (0.30-7.75)	0.61 (0.17-2.19)	2.24	1.92	0.14	0.11
TLR-guided	7.13	15.42	0.14 (0.04-0.44)	0.20 (0.08-0.54)	1.49	1.30	-0.61	-0.51
TMB/BRAF- guided	13.12	23.01	0.77 (0.29-2.05)	0.90 (0.39-2.11)	2.12	1.82	0.02	0.02

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